

Prayas JEE 2025

Maths

DPP: 5

Trigonometric Functions

Q1 Proof :

$$\frac{\tan 5\theta + \tan 3\theta}{\tan 5\theta - \tan 3\theta} = 4 \cos 2\theta \cos 4\theta.$$

Q2 Proof :

$$\frac{\cos 3\theta + 2 \cos 5\theta + \cos 7\theta}{\cos \theta + 2 \cos 3\theta + \cos 5\theta} = \cos 2\theta - \sin 2\theta \tan 3\theta.$$

Q3 Proof :

$$\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A.$$

Q4 The value of $\sin \frac{\pi}{18} + \sin \frac{\pi}{9} + \sin \frac{2\pi}{9} + \sin \frac{5\pi}{18}$ is

- (A) $\sin \frac{7\pi}{18} + \sin \frac{4\pi}{9}$
 (B) 1
 (C) $\cos \frac{\pi}{6} + \cos \frac{3\pi}{7}$
 (D) $\cos \frac{\pi}{9} + \sin \frac{\pi}{9}$

Q5 The value of

$$2 \sin^2 \theta + 4 \cos(\theta + \alpha) \sin \alpha \sin \theta + \cos 2(\alpha + \theta)$$

- is
 (A) $\cos \theta + \cos \alpha$
 (B) independent of θ
 (C) independent of α
 (D) independent of both θ and α

Q6 The value of $2 \tan 18^\circ + 3 \sec 18^\circ - 4 \cos 18^\circ$ is

- (A) Zero

(B) $\sqrt{5}$

(C) $-\sqrt{5}$

(D) $\sqrt{3}$

Q7 Let $\frac{3\pi}{4} < \theta < \pi$ and $\sqrt{2 \cot \theta + \frac{1}{\sin^2 \theta}} = k - \cot \theta$, then k is equal to

- (A) 1
 (B) -1
 (C) 0
 (D) $1/2$

Q8 The numerical value of $\sin \frac{\pi}{18} \sin \frac{5\pi}{18} \sin \frac{7\pi}{18}$ is equal to

- (A) 1
 (B) $1/8$
 (C) $1/4$
 (D) $1/4$

Q9 If

$$\tan \theta + \tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta - \frac{\pi}{3}\right) = k \tan 3\theta,$$

then k is equal to

- (A) 1
 (B) 3
 (C) $1/3$
 (D) none of these

Q10 The expression $\frac{\cos 6x + 6 \cos 4x + 15 \cos 2x + 10}{\cos 5x + 5 \cos 3x + 10 \cos x}$ is equal to

- (A) $\cos^2 x$
 (B) $1 + \cos x$
 (C) $\cos 2x$
 (D) $2 \cos x$



Answer Key

Q1 Proof

Q2 proof

Q3 proof

Q4 (A)

Q5 (B)

Q6 (A)

Q7 (B)

Q8 (B)

Q9 (B)

Q10 (D)



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